BINF*6210 – University of Guelph



T-Rex (Trichinella Repeat Explorer) Project

COMPARING THE TRANSPOSABLE ELEMENTS IN *T. SPIRALIS* & *T. MURRELLI*

Looking at Repeat Composition, Sequence Divergence, and Contig Distribution Across Two Related Parasites.

By: Hasan, Anusha, Hala, Yumai, and Moses

Shared Biology & Muscle Capsules

- **Muscle capsules:** Both parasites live inside host muscle cells surrounded by a protective collagen capsule (Murrell et al., 2000).
- **Same appearance:** Both species look almost identical under a microscope (Murrell et al., 2000)
- **Evolutionary split:** Trichinella species separated within the last 20 million years as they spread to new regions and hosts (Zarlenga et al., 2006)
- **Why compare them?** Their shared biology gives us a strong starting point to compare their repetitive DNA.

Evolutionary Timescale

~ 20 MYA

When modern Trichinella species began splitting as they spread across continents and hosts (Zarlenga et al., 2006)

Introduction – Host Parasitic Interactions

Different Hosts & Infection Patterns

	Trichinella SPIRALIS	Trichinella MURRELLI
MAIN HOST	Domestic Pigs and Rats around farms (Murrell et al., 2000)	Wild carnivores across North America (Pozio & La Rosa, 2000)
INFECTION RATE	High Pig infection (in experimentally infected pigs) (Wang et al., 2020)	Poor in Domestic Pigs (Wang et al., 2020)
IMMUNOLOGICAL RESPONSE	Triggers strong Antibody response during muscle infection (Wang et al., 2020)	Adapted for survival in wild hosts rather than farm animals (Pozio & La Rosa, 2000)

How They Were Identified as Separate Species

Step 1 – Before 2000

Grouped as Genotype T5

Similar larvae phenotype between the two species led to classification of *T. murrelli* as Genotype T5 (Murrell et al., 2000)

Step 2 – Year 2000

Named a Separate Species

Identification as separate species due to unsuccessful breeding tests between *T. spiralis* and *T. murrelli* (Pozio & La Rosa, 2000)

Step 3 – Year 2017

DNA Marker Differences

T. murrelli can show an extra 256-bp PCR band from its repetitive ITS1 region, which can confuse lab tests (Pozio & La Rosa, 2000)

Same Gene Order, Different DNA Sequence

- **Same gene order:** both species have the same mitochondrial genes in the same order (Webb & Rosenthal, 2011).
- **Uneven Changes:** some individual genes changed much more than others over time (Webb & Rosenthal, 2011).
- **Important Limit:** This 9.5% difference is only for mitochondrial DNA, so we still need to test nuclear transposable elements directly.

9.5%

Mitochondrial DNA sequence difference between T. spiralis and T. murrelli (Webb & Rosenthal, 2011).

Shared Gene Order & Genome Assembly Updates

- **Conserved gene order (synteny):** *T. spiralis* and *T. murrelli* maintained very similar gene order across their chromosomes even after 7 million years apart (Thompson et al. 2017).
- **Better repeat mapping:** long-read sequencing of *T. murrelli* filled in about 8 Mb of repetitive DNA that was missing before (Thompson et al., 2017).
- **Fixed old errors:** this study also found assembly mistakes in the older *T. spiralis* reference genome (Thompson et al., 2017).
- **Why it matters:** since their overall gene order stayed so similar, we want to see if their transposable elements also stayed the same.

Repetitive DNA Across the Genus

6.7 - 21.8%

Repeats make up 6.7% to 21.8% of 16 Trichinella genome assemblies, including DNA transposons and retrotransposons (Korhonen at al., 2016).

18% Repeats

Repeats make up ~18% of the 64-Mb *T. spiralis* genome and have lower GC content (27%) than coding genes (43%) (Mitreva et al., 2011)

15.3 vs. 4.3 Mb

T. pseudospiralis has much more LTR retrotransposon DNA (15.31 Mb) than *T. spiralis* (4.33 Mb),which has been linked to gene regulation (Liu et al., 2021).

Introduction – Biological Research Question

Research Question & Hypothesis

"Even though *T. spiralis* and *T. murrelli* share very similar gene order, do they exhibit differences in repetitive DNA composition, sequence divergence, and contig-level distribution that may reflect distinct evolutionary histories?"

- **By T-REX Team**

Alternative Hypothesis (H1)

Despite substantial genomic synteny, *T.spiralis* and *T.murrelli* may exhibit differences in specific repetitive DNA characteristics due to their distinct evolutionary histories.

Null Hypothesis (Ho)

There is no significant difference in repetitive DNA composition, sequence divergence, or contig-level distribution between *T. spiralis* and *T. murrelli.*

Methods – Data Access, Wrangling, and Filtering

```r
####Data Access----
##NCBI link for T. Spirallis and T. Murrelli
#T. Spirallis: https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_001447595.1/
#T. Murrelli: https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_001447425.1/

####Installation----
#Package installation & load libraries + Data exploration for both species (T. spiralis & T. murrelli)

install.packages("tidyverse")
install.packages("permute")
install.packages("patchwork")

rm(list = ls())

library(tidyverse)
conflicted::conflict_prefer("filter", "dplyr")
library(ggplot2)
library(dplyr)
library(stringr)
library(patchwork)

###2 - DATA EXPLORATION----

# Data wrangling. We will add a species label to each dataset and then combine them into one master table.

# Date of Download: October 23, 2026

#####trichinella spiralis----
spiralis_raw <- read_csv("../data/trichinella_spiralis_te.csv")

# check class
class(spiralis_raw)

# see variables
names(spiralis_raw)

# see summary
summary(spiralis_raw)

# check dimensions
dim(spiralis_raw)

#####trichinella murelli----
murrelli_raw <- read_csv("../data/trichinella murrelli_te.csv")

# check class
class(murrelli_raw)

# see variables
names(murrelli_raw)

# see summary
summary(murrelli_raw)

# check dimensions
dim(murrelli_raw)
```

- **Data Access (readr):** Imported NCBI RepeatMasker CSVs for *T. spiralis* and *T. murrelli* (read_csv()) and checked table structure (dim(), summary()).
- **Table Merging (dplyr):** Selected core variables (qry_id, qry_width, matching_class, perc_div, sw_score), tagged species names, and combined tables with bind_rows().

- **Broad Classification (stringr):** Used case_when() and str_detect() to group detailed annotations into major classes (LTR, PLE, DNA Transposon, LINE, SINE, RC/Helitron, Satellite).
- **Filtering & Length (dplyr):** Filtered out Simple Repeat, Low Complexity, and Other/Unknown rows (filter()) and calculated sequence length (qry_end - qry_start + 1)

```r
###3 - DATA CLEANING----
#####Combination----

spiralis_clean <- spiralis_raw |>
  select(qry_id, qry_start, qry_end, qry_width, matching_class, perc_div, sw_score) |>
  mutate(species = "T. spiralis")

murrelli_clean <- murrelli_raw |>
  select(qry_id, qry_start, qry_end, qry_width, matching_class, perc_div, sw_score) |>
  mutate(species = "T. murrelli")

df_tstm <- bind_rows(spiralis_clean, murrelli_clean)

####Broad Classification----
# Group detailed repeat annotations into broad, interpretable classes using str_detect() from stingr.

unique(df_tstm$matching_class)

df_tstm <- df_tstm |>
  mutate(
    broad_class = case_when(
      str_detect(matching_class, "DNA") ~ "DNA Transposon",
      str_detect(matching_class, "LINE") ~ "LINE",
      str_detect(matching_class, "SINE") ~ "SINE",
      str_detect(matching_class, "LTR") ~ "LTR Retrotransposon",
      str_detect(matching_class, "Simple_repeat") ~ "Simple Repeat",
      str_detect(matching_class, "simple_repeat") ~ "Simple Repeat",
      str_detect(matching_class, "Satellite") ~ "Satellite",
      str_detect(matching_class, "PLE") ~ "PLE",
      str_detect(matching_class, "Helitron") ~ "RC/Helitron",
      str_detect(matching_class, "Low_complexity") ~ "Low Complexity",
      str_detect(matching_class, "Retroposon") ~ "Other Retrotransposon",
      TRUE ~ "Other/Unknown"))

#check unique groups in broad_class
unique(df_tstm$broad_class)

####Non-TE Filtering----

# Filter out non-TEs

combined_te <- df_tstm |>
  filter(!broad_class %in% c("Simple Repeat", "Low Complexity", "Other/Unknown")) |>
  mutate(broad_class = fct_drop(broad_class)) |>
  mutate(broad_class = fct_infreq(broad_class))

df_tstmclean <- combined_te |>
  mutate(df_telength = qry_end - qry_start +1)

df_tstmclean |>
  select(qry_start, qry_end, df_telength) |>
  head()
```

Methods – Statistical Analysis & Figure Code

Figure 1 – Contig Density (stats, ggplot2): Counted repeats per contig (count(species, qry_id)), ran a Wilcoxon test (wilcox.test()), and plotted log10 boxplots (geom_boxplot()).

Figure 2 – Divergence Dumbbell Plots (dplyr, patchwork): Split subtypes at 5% divergence (perc_div >= 5 vs. <= 5), averaged divergence by subtype (mean()), and combined dumbbell plots.

Figure 3 – Multivariable Bubble Plots (ggplot2, patchwork): Plotted perc_div vs. df_telength scaled by sw_score, comparing all retained classes against a subset excluding LTR Retrotransposon.

```r
####Contig repeat density----

#step 8: figure 3: Contig-Level Repeat Density & Wilcoxon Test (Novelty Component)
#count number of repeats per contig using qry_id

contig_te_counts <- combined_te |>
  count(species, qry_id)

#perform Wilcoxon rank-sum test

wilcox_te_res <- wilcox.test(n ~ species, data = contig_te_counts)
p_val_label <- paste0("Wilcoxon rank-sum p = ", format.pval(wilcox_te_res$p.value, digits = 3))

fig3 <- ggplot(contig_te_counts, aes(x = species, y = n, fill = species)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.2) +
  scale_y_log10() +
  scale_fill_manual(values = c("T. spiralis" = "#2b8cbe", "T. murrelli" = "#e6550d")) +
  annotate("text", x = 1.5, y = max(contig_te_counts$n) * 0.8, label = p_val_label, fontface = "italic") +
  labs(
    title = "Transposable Element Density per Contig",
    subtitle = "Comparing spatial TE clustering across assembly contigs",
    x = "Species",
    y = "Number of TEs per Contig (log10 scale)"
  ) +
  theme_classic() +
  theme(legend.position = "none")

print(fig3)
ggsave("../output/figure3_contig_density.png", plot = fig3, width = 6, height = 4.5)
```

Figure 1 (Contig-Level Repeat Density)

Repeat Annotations Per Contig Diverge Significantly Between Species

- **More spread out in *T. murrelli*:** has a median of 2 TEs per contig (box from 1 to 3, whisker to ~12), showing TEs are spread across many contigs.
- **Crowded contigs in *T. spiralis*:** has a lower median (1 TE per contig) but several extreme outlier contigs with over 100 TEs.
- **Statistical Result:** the Wilcoxon test (p = 6.17e- 14) shows a clear difference in how TEs are grouped across contigs.

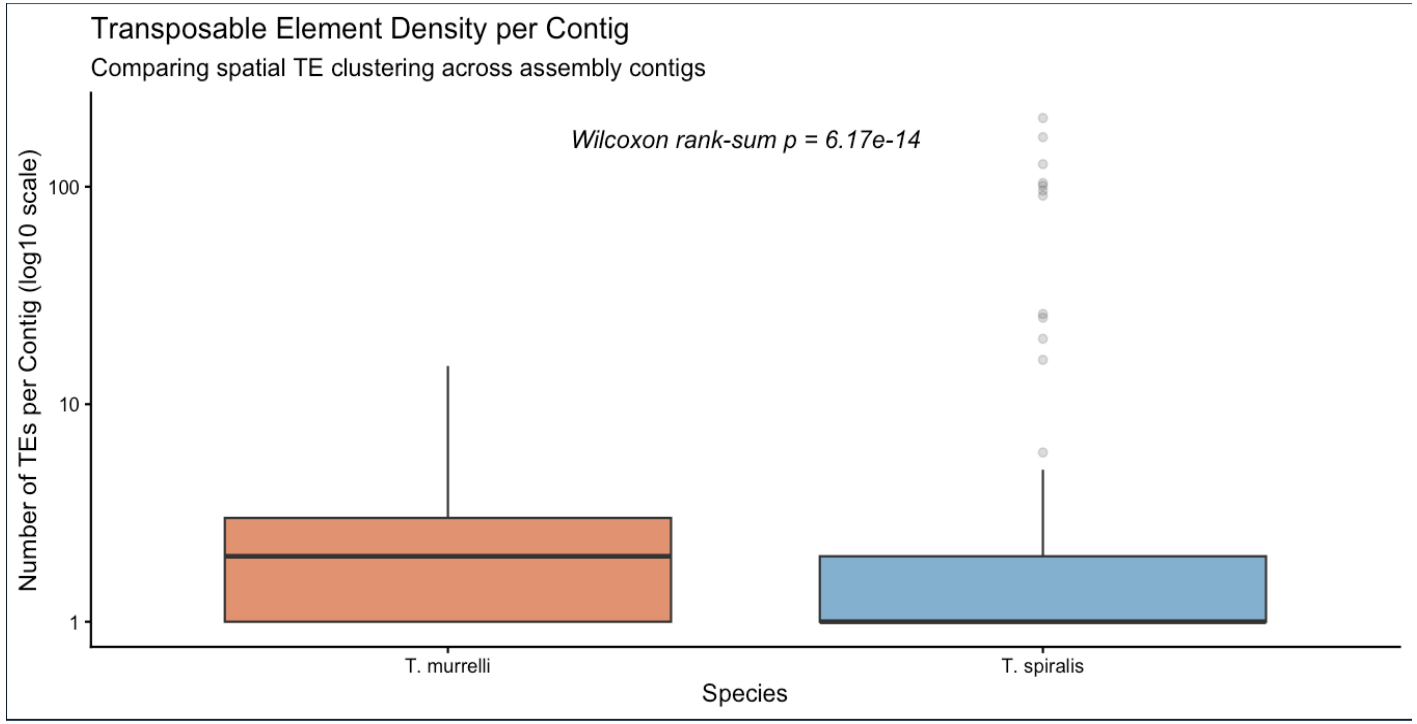
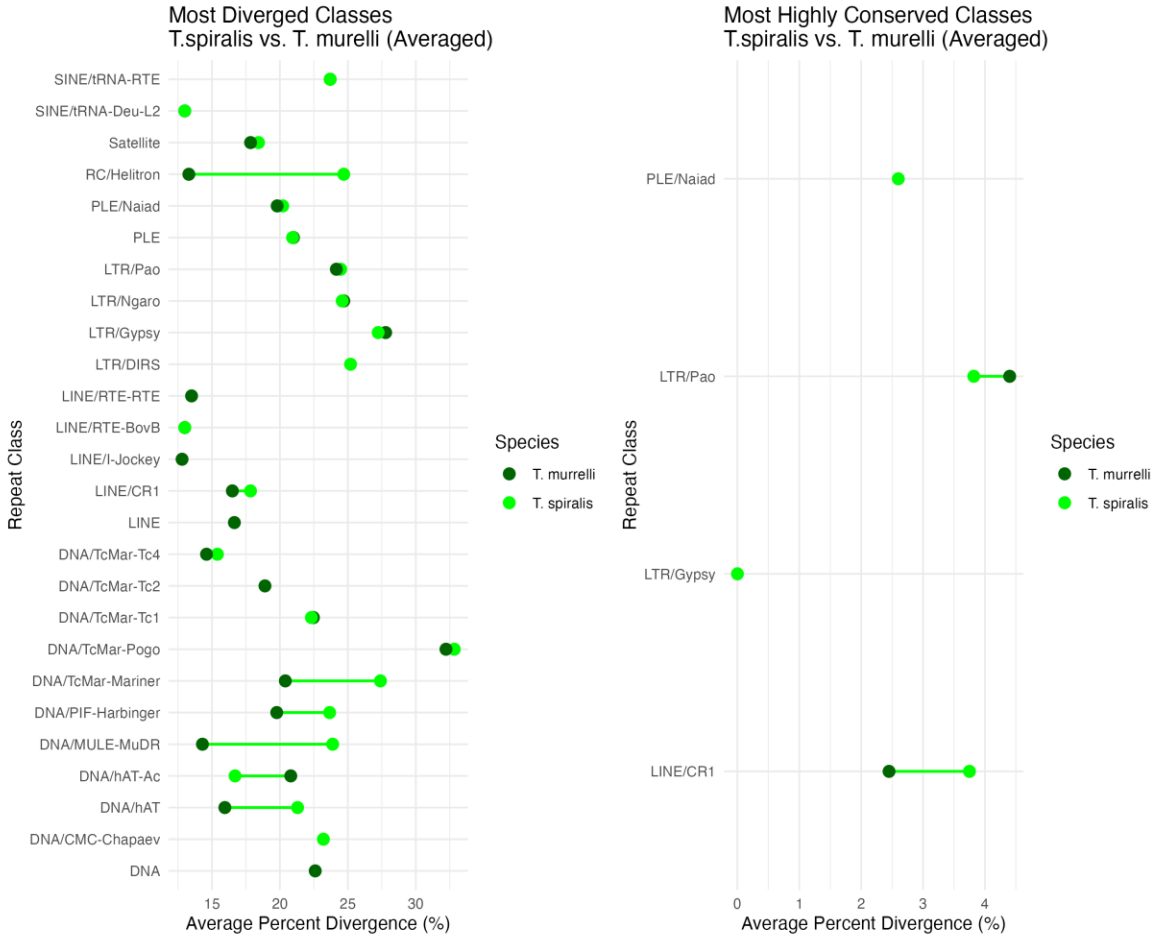


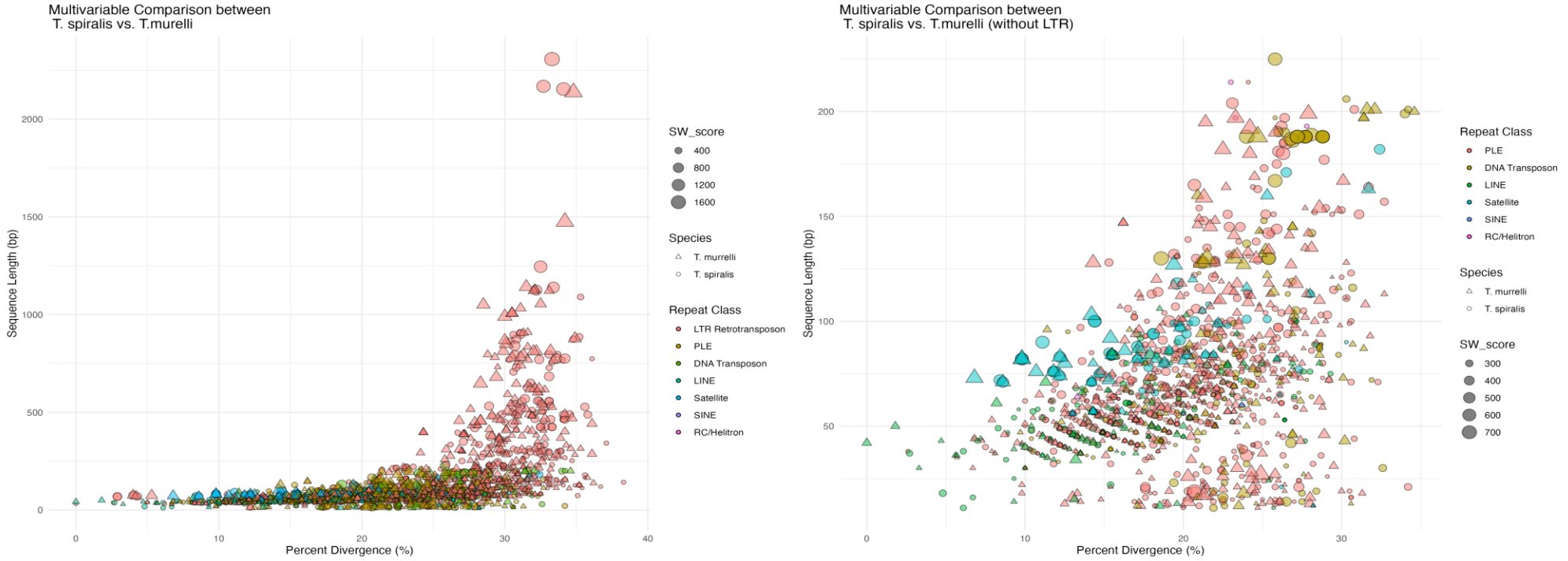
Figure 2 (Dumbbell Plot: Diverged vs. Conserved Subtypes)

Specific DNA Transposon and Helitron Subtypes are more Conserved in *T. murrelli* than in *T.spiralis*



- **5% Divergence Cutoff:** Subtypes were split at 5% sequence divergence (referenced to the kimura divergence cut-off to compare more mutated (=/> 5%) vs. Highly conserved (=/< 5%) repeats (Palacios-Gimenez, 2020).
- **Higher Divergence in T. spiralis:** Interestingly, large variation in divergence is observed for RC/Helitron and DNA/MULE-MuDR, demonstrating *T.spiralis* appears to have diverged much more from the consensus *T.murrelli*, which is hypothesized to have diverged later.
- **Strong Conservation of LTR/Pao and LINE/CR1:** Strong conservation of these repeats between the two species could hint at critical functional impact
- **Biological Link:** More stable DNA transposon and Helitron sequences in *T. murrelli* may support adaptation to wild hosts over farm pigs.

Figure 3 (Multivariable Comparison)
Removing Dominant LTRs Reveals Short, Highly Mutated DNA Transposon and PLE Fragments



- **LTR Dominance (Left Plot):** LTR Retrotransposons are the longest repeats (up to ~2,000 bp) with high divergence (25–35%) and high intactness (SW_score).
- **Zooming In (Right Plot):** Removing LTRs rescales the y-axis (<250 bp) so shorter repeat classes are clearly visible.
- **Short, Diverged Repeats:** Many DNA Transposon and PLE copies show higher divergence (20–35%) despite short lengths (<150 bp) and lower SW_score values (fragmented copies).
- **Evolutionary Value:** Short, highly mutated fragments and high-scoring outliers point to older TE insertions worth checking near functional genes. Investigation into increasing resolution (subtype) for outliers could pose interesting future avenue of study.

Results & Discussion (Biological Interpretation of Findings)

Related species retain similar repeat profiles despite sequence-level variation

Major similarities:

- LTR retrotransposons and PLEs dominate the retained annotations in both species, together representing approximately 77-80% of records.
- Median repeat fragment lengths are nearly identical: 76 bp in *T.spiralis* and 75 bp in *T.murrelli*
- Satellite repeats also occur in similar proportions(4.30% and 4.23%), suggesting comparable representation of this non-TE category.

Major Differences:

- *T.murrelli* has a slightly higher proportion of LTR retrotransposons (44.63% vs 41.81%) whereas *T.spiralis* has a higher proportion of DNA transposons (9.81% vs 7.92%)
- Sequence divergence varies across TE classes, with LTR retrotransposons exhibiting relatively high divergence and LINEs showing lower median divergence.
- Repeat annotation counts per contig differ between species although assembly characteristics may contribute too this pattern.

Biological Interpretation:

- Despite millions of years of evolutionary divergence, both species retain broadly similar repetitive DNA profiles. These findings are consistent with their shared ancestry and substantial genome synteny, while differences within repeat classes may reflect distinct evolutionary histories (Thompson et al., 2017; Korhonen et al., 2016)

Results & Discussion (Limitations and Data Biases)

Assembly and annotation limitations affect interpretation of repeat variation

- **Genome assembly differences:** Contig lengths and assembly fragmentation may influence the observed distribution of repeat annotations (Thompson et al., 2017).
- **Annotation fragmentation:** A single TE insertion may generate multiple RepeatMasker records, meaning annotation counts do not necessarily represent complete TE copies (Baril et al., 2024).
- **Contig-level abundance:** Raw counts do not account for contig length and cannot establish differences in actual repeat density or spatial clustering.
- **Sequence divergence:** Percent divergence is measured relative to repeat consensus sequences, rather than between corresponding genomic regions in the two species.
- **Satellite repeats:** Similar annotation proportions do not demonstrate conservation of satellite families, repeat arrays, or genomic positions.
- **Evolutionary interpretation:** The present analysis cannot determine historical insertion timing or whether repeat differences contributed to host adaptation.

Results & Discussion (Future directions and next steps)

TE similarities coexist with differences in annotation distribution

Overall conclusions:

- *T.spiralis* and *T.murrelli* share broadly similar repetitive element compositions, including comparable proportions of LTR retrotransposons, PLEs and satellite repeats.
- Repeat fragment lengths and major TE divergence patterns are generally similar between species, although individual repeat class exhibit variation.
- The distributions of repeat annotation counts per contig differ, but these differences cannot yet be separated from potential assembly and annotation effects.
- Overall, our results show that substantial genome synteny can coexist with variation in repetitive DNA characteristics, without establishing the evolutionary mechanisms responsible.

Future directions:

- Normalize repeat abundance by contig length and compare repeat coverage in base pairs
- Investigate individual TE families to identify class-specific differences in abundance and divergence
- Compare Satellite repeat families and their genomic positions to determine whether their similar overall proportions conceal sequence-level differences.
- Use improved genome assemblies and comparative alignments to identify shared and species-specific TE insertions.

References

- Baril, T., Galbraith, J., & Hayward, A. (2024). Earl Grey: A fully automated user-friendly transposable element annotation and analysis pipeline. *Molecular Biology and Evolution*, *41*(4), msae068. https://doi.org/10.1093/molbev/msae068
- Karadjian, G., Heckmann, A., & Vallée, I. (2017). Molecular identification of *Trichinella* species: An unexpected PCR pattern for *Trichinella murrelli*. *Parasites & Vectors*, *10*(1), 606. https://doi.org/10.1186/s13071-017-2550-y
- Korhonen, P. K., Pozio, E., La Rosa, G., Chang, B. C., Koehler, A. V., Hoberg, E. P., Boag, P. R., Tan, P., Jex, A. R., Hofmann, A., Sternberg, P. W., & Gasser, R. B. (2016). Phylogenomic and biogeographic reconstruction of the *Trichinella* complex. *Nature Communications*, *7*, 10565. https://doi.org/10.1038/ncomms10565
- Liu, X., Wang, Z., Bai, X., Wang, X., Tang, B., Guan, Z., Song, Y., Liu, C., & Liu, M. (2021). Comparative multi-omics analyses reveal differential expression of repetitive elements and parasitism genes between encapsulated and non-encapsulated *Trichinella* species. *Nucleic Acids Research*, *49*(2), 700–718. https://doi.org/10.1093/nar/gkaa1218
- Mitreva, M., Jasmer, D. P., Zarlenga, D. S., Wang, Z., Abubucker, S., Martin, J., Taylor, C. M., Yin, Y., Fulton, L., Minx, P., Yang, S. P., Warren, W. C., Fulton, R. S., Bhonagiri, V., Zhang, X., Hallsworth-Pepin, K., Clifton, S. W., McCarter, J. P., & Wilson, R. K. (2011). The draft genome of the parasitic nematode *Trichinella spiralis*. *Nature Genetics*, *43*(3), 228–235. https://doi.org/10.1038/ng.769
- Murrell, K. D., Lichtenfels, J. R., Zarlenga, D. S., & Pozio, E. (2000). The systematics of the genus *Trichinella* with a key to species. *Veterinary Parasitology*, *93*(3–4), 293–307. https://doi.org/10.1016/S0304-4017(00)00347-1
- Palacios-Gimenez, O. M., Koelman, J., Palmada-Flores, M., Bradford, T. M., Jones, K. K., Cooper, S. J. B., Kawakami, T., & Suh, A. (2020). Comparative analysis of morabine grasshopper genomes reveals highly abundant transposable elements and rapidly proliferating satellite DNA repeats. *BMC biology*, *18*(1), 199. https://doi.org/10.1186/s12915-020-00925-x

References

- Pozio, E., & La Rosa, G. (2000). *Trichinella murrelli* n. sp: Etiological agent of sylvatic trichinellosis in temperate areas of North America. *The Journal of Parasitology*, *86*(1), 134–139. https://doi.org/10.1645/0022-3395(2000)086
- Thompson, P. C., Zarlenga, D. S., Liu, M.-Y., & Rosenthal, B. M. (2017). Long-read sequencing improves assembly of *Trichinella* genomes 10-fold, revealing substantial synteny between lineages diverged over 7 million years. *Parasitology*, *144*(10), 1302–1315. https://doi.org/10.1017/S0031182017000348
- Wang, Z. Q., Wang, L., Cui, J., & Jiang, P. (2020). Comparative infectivity and antibody response of four *Trichinella* species in experimentally infected swine. *Parasites & Vectors*, *13*(1), 209. https://doi.org/10.1186/s13071-020-04085-7
- Webb, K. M., & Rosenthal, B. M. (2011). Comparative mitochondrial genomics of *Trichinella* reveals conserved synteny with marked nucleotide sequence divergence. *Gene*, *473*(2), 161–168. https://doi.org/10.1016/j.gene.2010.11.015
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., … Yutani, H. (2019). Welcome to the Tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686
- Zarlenga, D. S., Chute, M. B., Martin, A., & Kapel, C. M. (2006). A multiplex PCR for unambiguous identification of all encapsulated and non-encapsulated genotypes of *Trichinella*. *International Journal for Parasitology*, *36*(7), 837–844. https://doi.org/10.1016/j.ijpara.2006.03.011

CoPilot Disclosure - Anusha

1. What are the major biological and evolutionary similarities and differences between Trichinella spiralis and Trichinella murrelli.
2. How could the conserved genome synteny between T.spiralis and T.murrelli help us develop a biologically relevant research question about repetitive DNA?
3. Can you suggest some peer-reviewed research papers discussing repetitive DNA, transposable elements and genome evolution in Trichinella species
4. What does sequence divergence from a repeat consensus tell us about transposable element evolution and what are its limitations
5. What biological factors could explain differences in repetitive element abundance and distribution between two closely related species and how might genome assembly quality affect these comparisons?
6. How can two species maintain similar transposable element compositions despite millions of years of evolutionary divergence and what biological processes might explain differences between them?
7. How do Satellite repeats differ from transposable elements, and what could similarities in their abundance suggest about genome evolution in closely related species.
8. Self reflection assignment: Check for any grammatical or spelling errors in this text.

CoPilot / Stack Overflow Disclosure - Yumai

- https://stackoverflow.com/questions/60675989/bubble-map-legend-color-in-ggplot2-using-viridis (troubleshoot legend colouring issue)
- https://stackoverflow.com/questions/73141343/how-to-arrange-categorical-bubble-plot-in-ggplot2 (bubble plot code explanation)

- Prompts:
  - What are some alternatives to scatter plots?
  - Can you recommend me a short list of papers comparing Trichinella spiralis and Trichinella murrelli?
  - Nematode percent divergence per million years and how is it calculated
  - Explain kimura divergence and calculate for trichinella
  - Transposable element Kimura divergence for trichinella. I want a percent divergence cut-off for likeness

CoPilot Disclosure - Hasan

- What is a good group name that will represent or show the transposable elements between different species using R studio?
- What is the difference between T. spiralis and T. murrelli?
- What is considered a TE vs non-TE?
- What is the best test to perform in RStudio to show the difference in data between two species?
- What is a Wilcoxon test and how can I perform it in RStudio?
- Rearrange this reference list in alphabetic order.
- Check for grammar mistakes in this writing.

CoPilot Disclosure - Hala

- What is the difference between T. Spiralis and T. Murelli
- What do these variables mean in terms of transposable elements from repeatmasker output: sw_score perc_div perc_del perc_insert qry_id qry_start qry_end qry_left matching_repeat repeat_id matching_class no_bp_in_complement in_repeat_start in_repeat_end qry_width
- What type of test can i do if data is not normally distributed
- What is a wilcoxon test
- Smaller than expected counts in one category for chi-square what to do?
- Fisher's test vs chi-square
- What kind of figures can be created to show divergence by class
- Correct grammar

CoPilot Disclosure - Moses

- Research materials, relevant academic resources, and information shared collaboratively among group members.